

NICOLAS ESSER (HE/HIM) - 2026 PHD GRADUATE

✉ nicolas.esser@ulb.be

🌐 Website: nicolasesser.github.io

📍 PhD @ Université libre de Bruxelles
Incoming Postdoc @ Yale University

🇧🇪 Belgian, 26 years old

INSPIRE

ORCID

KEY POINTS

- Theoretical physicist with observational skills; interested in dark matter & faint MW satellites
- 4 first-author, 2 single-author and 1 co-author publications during the PhD
- PhD thesis defended June 19, 2026
- Fulbright Fellow 2026-2027 @ Yale Department of Astronomy
- ~ 20 talks delivered; attended ~ 12 conferences and 3 doctoral schools
- 300+ hours of teaching, supervision, and outreach in the last 4 years

RESEARCH INTERESTS

- Ultra-faint dwarf galaxies & their dark matter content
- Stellar dynamics & dynamical friction
- Primordial black holes
- Dark matter phenomenology & interactions with stars
- Bayesian analysis; synthetic and observed stellar populations
- Strong background in theoretical particle physics

FIRST-AUTHOR PUBLICATIONS

4. Esser, García-Bellido & Tinyakov 2026

Esser, Nicolas; García-Bellido, Juan; Tinyakov, Peter. "Gravitational waves from primordial black holes passing by neutron stars: Observational prospects for the Galactic center".

[Physical Review D](#), Volume 113, 123053, June 2026, arXiv:2602.23429

3. Esser & al. 2025

Esser, Nicolas; Filion, Carrie; De Rijcke, Sven; Kallivayalil, Nitya; Richstein, Hannah; Tinyakov, Peter; Wyse, Rosemary F.G. "Constraints on asteroid-mass primordial black holes in dwarf galaxies using Hubble Space Telescope photometry".

[Astronomy & Astrophysics](#), Volume 698, A290, May 2025, arXiv:2503.03352

2. Esser, De Rijcke & Tinyakov 2024

Esser, Nicolas; De Rijcke, Sven; Tinyakov, Peter. "The impact of primordial black holes on the stellar mass function of ultra-faint dwarf galaxies".

[Monthly notices of the Royal Astronomical Society](#), Volume 529, Issue 1, Pages 32–40, March 2024, arXiv:2311.12658

1. Esser & Tinyakov 2023

Esser, Nicolas; Tinyakov, Peter. "Constraints on primordial black holes from observation of stars in dwarf galaxies".

[Physical Review D](#), Volume 107, 103052, May 2023, arXiv:2207.07412

SINGLE AUTHOR PUBLICATIONS

2. Esser 2025

Esser, Nicolas. "Primordial black hole-star binaries via dynamical friction".

[Physical Review D](#), Volume 112, August 2025, 043021, arXiv:2505.05564

1. Esser 2026

Esser, Nicolas. "Dynamical friction on binary stars in dark matter dominated environments".
[Arxiv preprint](#), arXiv:2602.20245

CO-AUTHOR PUBLICATIONS

1. Errani, Esser, Peñarrubia & Walker 2026

Errani, Raphaël; Esser, Nicolas; Peñarrubia Jorge; Walker, Matthew G. "Collisional Dynamics of Stars and Dark Matter in Ultra-Faint Galaxies".
[The Astrophysical Journal](#), Volume 1007, 42, August 2026, arXiv:2604.06304

PUBLICATIONS IN PREPARATION

3. Esser, Nicolas; Tinyakov Peter. "Accretion of stars by small black holes: the role of rotation".

2. Dandoy Virgile; Esser, Nicolas. "Axion miniclusters in dwarf galaxies".

1. Esser, Nicolas; Franciolini Gabriele. "Dynamical friction in primordial black hole binary formation".

ACADEMIC SUPERVISORS AND MENTORS

- **(Theory)** Prof. Peter Tinyakov — PhD Advisor  petr.tiniakov@ulb.be
Université libre de Bruxelles
- **(Astronomy)** Prof. Sven De Rijcke — Collaborator  sven.derijcke@ugent.be
Ghent University
- **(Theory)** Prof. Sébastien Clesse — Thesis advisory committee  sebastien.clesse@ulb.be
Université libre de Bruxelles
- **(Astronomy)** Prof. Earl Bellinger — Postdoc Advisor  earl.bellinger@yale.edu
Yale University
- **(Astronomy)** Prof. Marla Geha — Postdoc Advisor  marla.geha@yale.edu
Yale University

EDUCATION

- Oct. 2022 – June 2026 **PhD in Physics** – Université libre de Bruxelles
Primordial black holes & stars in ultra-faint dwarf galaxies
- Sept. 2020 – June 2022 **M.Sc. in Physics** – Université libre de Bruxelles Grade: 17.3/20
Major: theoretical physics; minor: stellar astrophysics
- Sept. 2017– June 2020 **B.Sc. in Physics** – Université libre de Bruxelles Grade: 15.5/20

GRANTS

 2026 **Fulbright research scholarship**
Postdoctoral fellow 2026-2027 @ Yale Astronomy

 2026 **Belgian American Educational Foundation (BAEF) research fellowship**
Postdoctoral fellow 2026-2027 @ Yale Astronomy

 2023 **FRIA PhD research grant**
Selected by the FNRS (Belgian national research agency)

 2022 **Fully funded teaching assistant PhD position**
Left the position early upon obtaining the FRIA research grant

AWARDS

🏆 2024 **Audience award**, 'My thesis in 180 seconds' science communication contest
At the Belgian finals

🏆 2022 **AScBr prize**
For the best master thesis in natural sciences (first prize)

CONFERENCES, TALKS, AND SEMINARS

~ 20 research presentations (invited, contributed, seminar, and poster formats).

Selected recent highlights:

- **Poster + Contributed Talk** – Dark Matter and Stars 2026 Southampton, UK, July 2026
- **Invited Seminar Talk** – Yale Departement of Astronomy New Haven, USA, Nov. 2025
- **Invited Talk** – News From the Dark 10 Montpellier, France, Sept. 2025
- **Poster + Flash Talk** – Invisibles25 @ CERN Geneva, Switzerland, Sept. 2025
- **Invited Seminar Talk** – Max Planck Institute for Astronomy Heidelberg, Germany, August 2025
- **Contributed Talk** – European Astronomical Society Annual Meeting Cork, Ireland, June 2025

A full list of my presentations is available on my [website](#).

PROFESSIONAL EXPERIENCE

- **Postdoctoral Fellow** – Yale University Oct. 2026 – Expected Sept. 2027
Supervised by Prof. Earl Bellinger & Prof. Marla Geha. [Unveiling the nature and dark matter content of the faintest Milky Way satellites using binary stars.](#)
- **Graduate Researcher (PhD candidate)** – Université libre de Bruxelles Oct. 2022 – June 2026
Supervised by Prof. Peter Tinyakov. [Investigating the interactions of primordial black holes with stars and their potential observable signatures in ultra-faint dwarf galaxies.](#)
- **Graduate Researcher (Master student)** – Université libre de Bruxelles Sept. 2021 – June 2022
Supervised by Prof. Peter Tinyakov. [Research on the dynamics of primordial black hole capture by stars.](#)
- **Graduate Research Intern** – Université libre de Bruxelles Jan. – June 2021
Supervised by Prof. Geoffrey Compère. [Working on black hole thermodynamics.](#)
- **Undergraduate Research Intern** – Université libre de Bruxelles Jan. – June 2020
Supervised by Prof. Nicolas Chamel. [Testing the impact of Gaussian noise on simulated neutron star crust structures.](#)

TEACHING & MENTORING

250+ hours of teaching and supervision during my PhD. Details:

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|---|-------------|
| ✍ Supervised research intern Nour Mouhssine | 2024 – 2025 |
| <i>Bachelor thesis on 'A simple model of dark matter-star encounters'</i> | |
| ✍ Teaching assistant | 2023 – 2026 |
| <i>Assistant for general physics and special relativity, 4 hours/week</i> | |
| ✍ Teaching assistant | 2022 – 2023 |
| <i>Assistant for general physics, 8 hours/week</i> | |
| ✍ Lab assistant | 2022 – 2023 |
| <i>Mechanics & electricity, 4 hours/week</i> | |
| ✍ General physics tutor | 2021 – 2022 |

PROFESSIONAL AND DEPARTMENTAL SERVICE

 **Member of Local Organising Committee** – Massachusetts Institute of Technology
Dark Matter and Stars 2027, upcoming 12–14 July 2027

 **Member of Organising Committee (LOC + SOC)** – Yale Department of Astronomy
Tinsley Workshop: The Threshold of Galaxy Formation, upcoming 8–10 March 2027

 **Main organiser** – Université libre de Bruxelles
Annual BLU-Panda PhD day, for all theoretical and astrophysics students, since 2023

 **Co-chair of Local Organising Committee** – Université libre de Bruxelles
Conference: *New Horizons in Primordial Black Hole Physics 3*, 19–22 May 2025

 **Member of Local Organising Committee** – Université libre de Bruxelles
Workshop: *News from the Dark 8*, 11–13 September 2023

 **Peer reviewer**
Journal of Cosmology and Astroparticle Physics (JCAP) (since 2025) & The Astrophysical Journal (ApJ) (since 2026) & ESO DPR P117

OUTREACH

I consider outreach to be a fundamental part of the researcher's role. Recent highlights include:

- **Brussels public lectures of astronomy** 2026
Presented 4 public lectures on dark matter as part of the historic general public astronomy lectures of Brussels, held since 1823
- **Université des enfants** (children's university) 2023 – 2026
Monthly Saturday morning workshops for 6–12 year olds on electricity
- **'Qui a mangé le soleil ?' (Who ate the sun?)** interactive conference 2024 – 2025
Interactive conference on space & cosmology for children, with Prof. Sébastien Clesse, Dr. Jean Kimus & Dr. Léa Planquart
- **My thesis in 180 seconds** – Belgian qualifiers and finals 2024
Audience award winner
- **FNRS (Belgian national research agency) annual report** 2024
Produced an audio capsule
- **Invited on Canadian radio** – Quirks & Quarks, CBC radio one 2024
Presented my recent research on primordial black holes to a national audience

More details on my outreach activities are available on my [website](#).

LANGUAGES

 **Programming:** Python (Proficient) C/C++ Fortran Mathematica

 **Natural:** French (Native) English (Proficient) Dutch (Intermediate)